THE LOD & FORMULA DECODER CARD

Companion drill card for advanced-paper4.pdf - 60 seconds per question, no visualization needed

1. THE 30-SECOND LOD TRACE (do this on scratch paper)

Step 1 - INNER braces = draw the bar chart. {FIXED [City] : SUM([Sales])} means "a bar chart of Sales per City": Boston = 160, Austin = 80. The inner LOD is just that imaginary chart - nothing else.

Step 2 - OUTER wrapper = the verb on those bars. MAX = biggest bar | AVG = average bar | SUM = add all the bars (equals the level above) | COUNT = how many bars. So AVG({FIXED [City] : SUM([Sales])}) = "average of the city totals."

Step 3 - TIMING = which filters already fired? Locate the question's filter on the funnel in section 4. Fired before your LOD's stage -> the number changes. Fired after -> the number is frozen. Most wrong answers in the exam are only wrong here.

2. ONE-WORD PERSONALITIES (the visualization trick)

LOD	Personality	Mental image
{FIXED [dim] : X}	PHOTOCOPIER	Computes its summary ONCE and stamps it onto matching rows. Deaf to dimension and measure filters. Only Extract / Data-source / Context filters (stages ABOVE it) can change it. Can be saved as a dimension (e.g., customer's first order date).
{INCLUDE [dim] : X}	ZOOM-IN	Drops BELOW the view grain (Region view -> zoom to State), computes at the finer grain, aggregates back up. Wrong when the named dim is coarser than the view.
{EXCLUDE [dim] : X}	DELETE-A-COL	Cross one dimension off the view and compute as if it isn't there. Use for "compare each row to its parent total."
LOD inside LOD	ONION	Peel inside-out. Inner builds a per-row column first; outer summarizes it. Never read the outer braces first.
{FIXED : X} (no dim)	GRAND-TOTAL STAMP	One company-wide number broadcast onto every mark. "Same value everywhere" in the options = this.
Bare {X}	VIEW-LEVEL AGGREGATE	Same as the plain aggregate at the viz grain. A distractor 90% of the time.

3. KEYWORD -> ANSWER RECOGNITION (spot it before computing)

The stem says...	The correct answer contains...
"regardless of what's shown / even when the user filters"	{FIXED ...}
"share of the FULL / unfiltered total"	SUM(x) / {FIXED : SUM(x)} (FIXED denominator)
"compare each mark to its REGION / parent total"	{EXCLUDE [finer dim]}
"average per ORDER (not per line) / per CUSTOMER (not per row)"	AVG({INCLUDE [finer dim] : SUM(x)})
"first / last / most recent date per customer" as a label	{FIXED [Customer] : MIN/MAX([Order Date])}
"bucket/rank customers by lifetime value"	FIXED saved as a dimension, then CASE/IF bands
"same number on every mark / grand total everywhere"	{FIXED : SUM(x)}

4. THE FILTER FUNNEL (memorize the ladder)

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Extract -> Data source -> Context -> Sets/Top-N -> FIXED -> Dimension -> INCLUDE/EXCLUDE -> Measure -> Table-calc filter
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"FIXED is deaf to dimension filters - only Context (and above) can shout at it. INCLUDE/EXCLUDE hear everything." A table-calc filter hides marks only; it never changes FIXED, totals, or % -of-total denominators.

"Top 10 WITHIN the selected category" -> promote the category filter to Context, so it fires before the Top-N stage.

5. DISTRACTOR RADAR (recognize the answer without full computation)

In LOD trace questions the wrong options are rarely random - they are the SAME number computed at a different moment or grain:

Trap option is...	It computed...
The raw row-level number	Grain too low (you forgot the LOD groups first)
The parent / grand total	Grain too high (you removed a level that braces keep)
The after-filter number	Timing error (filter fires AFTER the LOD's stage)
The SUM twin vs AVG twin	Right inner LOD, wrong OUTER wrapper

If two options are twins across one grain or one wrapper - THAT is the test. Trust the braces, not the viz. Cover the options, compute your own value on the 2-4 given rows, then match.

6. FORMULA QUICK-RECOGNITION (the non-LOD half)

Function	5-second rule
DATEDIFF(part, start, end)	FENCE POSTS: count part-boundaries crossed (Sundays for 'week', 1sts for 'month') - NOT elapsed time. Jan 31 -> Feb 1 = 1 month boundary. Negative = start is after end.
DATEADD(part, n, date)	Keeps the day-of-month if it exists; clamps to month-end only when the target day is invalid. Jun 30 + 6 months = Dec 30 (not Dec 31). Jan 31 + 1 month = Feb 28/29.
DATEPART / DATENAME / DATETRUNC	number (10) / word ("October") / first moment of the period ('month' -> #Oct 1#). Trunc never rounds up.
MAKEDATE / MAKEDATETIME / DATEPARSE	y,m,d -> date | date + time -> datetime | (format, string) -> parse text into a date. ISDATE tests validity, returns TRUE/FALSE.
LEFT / RIGHT / MID / SPLIT	Count characters from the named edge; positions start at 1, not 0. MID(s, start, length). SPLIT(s, "-", -1) takes the token from the END. CONTAINS is case-sensitive.
FLOOR / CEILING / INT / ROUND	Negatives: draw the number line -10 <-- -9.4 --> -9. FLOOR walks LEFT (-10), CEILING walks RIGHT (-9), INT walks TOWARD ZERO (-9), ROUND = nearest, .5 away from zero.
DIV(17, 5)	Integer division: 3 (remainder dropped, never rounded).
Logic	AND binds tighter than OR. Any comparison touching NULL = FALSE -> runs the ELSE. IIF(test, then, else[, unknown]) is a function; IF...THEN...END is a statement - don't mix the syntaxes.
ZN / IFNULL / ISNULL	ZN = numeric fields only, NULL -> 0. IFNULL = any type, any replacement. ISNULL = returns TRUE/FALSE only (a test, not a filler).
Ratios	Same grain top and bottom = SUM(x)/SUM(y). "Average of totals" = outer AVG around an LOD/INCLUDE. COUNTD for "distinct"; ATTR = * when a mark covers several values.

7. THE 30-SECOND EXAM SCRIPT

1. Cover the options with your hand. **2.** Inner braces -> draw the bar chart. **3.** Wrapper -> verb. **4.** Named filter -> stage it on the funnel: before or after the calc? **5.** Compute on the 2-4 given values, THEN match an option.

8. WORKED EXAMPLE (advanced paper 4, Q16)

`AVG({INCLUDE [State] : SUM([Sales])}) | view = Region level | East rows: NY 100, NY 100, NJ 300`

Step	Do	Result
1. Inner bar chart	Sales per State (zoom below the Region view)	NY = 200, NJ = 300
2. Wrapper verb	AVG over those bars	(200 + 300) / 2 = 250
3. Timing	No filter named -> nothing to check	answer = 250
Radar check	166.67 = raw row avg... 500 = SUM twin... 200 = NY alone	all three are trap twins - reject

Drill: redo Q13-Q22 of advanced-paper4.pdf using only this card. Target: 9/10 in under 3 minutes each. Then mark every LOD question you missed as grain, wrapper, or timing - whichever label repeats is the one to drill.